

ECON 0100 | Homework B4 | Efficiency

| Solutions

Answers and work are shown in the red blocks.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P_d = 100 - \frac{1}{2}Q_d$$

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q$$

Prices are in galleons and quantity is in bottles.

Q1 | Total Surplus

What is the market equilibrium price, quantity, and total surplus? Use a graph and algebra to analyze this market.

P*: _____

| 60 galleons.

| From Homework B3, $100 - \frac{1}{2}Q = 20 + \frac{1}{2}Q \rightarrow Q^* = 80$ and $P^* = 60$.

Q*: _____

| 80 bottles.

TS: _____

| 3200 galleons.

Total surplus is the whole triangle between demand and supply out to Q^* , with height the gap between the two intercepts:

$$TS = h \cdot b \cdot \frac{1}{2} = (100 - 20) \cdot 80 \cdot \frac{1}{2} = 3200 \quad (= CS + PS = 1600 + 1600)$$

This is the most surplus the market can generate: every bottle up to the eightieth has a buyer whose marginal benefit exceeds a seller's marginal cost, and none beyond it does.

Q2 | Price Ceiling

The Ministry of Magic decided to institute a price ceiling of 50 galleons to allow many more to afford the treat. Use a graph of the demand curve and the supply curve to evaluate the welfare effects this policy had on the market.

Quantity Supplied: _____

I 60 bottles.

Quantity Demanded: _____

I 100 bottles.

Excess, Market Clearing, or Shortage: _____

I Shortage.

CS: _____

I 2100 galleons.

How much did CS change? _____

I Up by 500 galleons.

PS: _____

I 900 galleons.

How much did PS change? _____

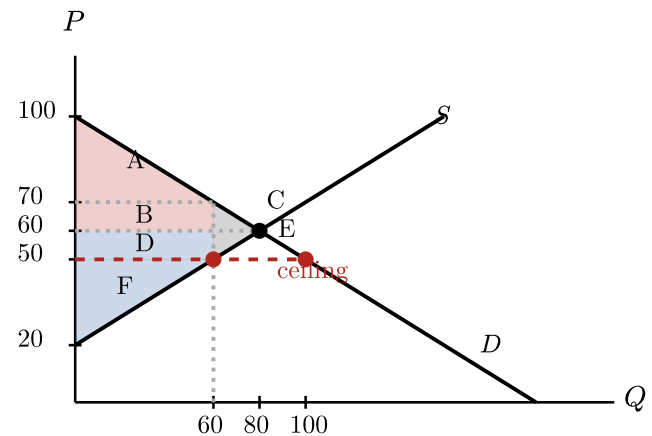
Down by 700 galleons.

DWL: _____

200 galleons.

The ceiling of 50 sits below $P^* = 60$, so it binds. At 50: $Q_s = 60$ and $Q_d = 100$, a shortage of 40 only 60 bottles are exchanged. The buyers who get them value the sixtieth bottle at $P_d = 100 - \frac{1}{2} \cdot 60 = 70$.

Let the regions between the curves out to $Q^* = 80$, split at $Q = 60$ and at the prices 70, 60, and 50:



$$A = (100 - 70) \cdot 60 \cdot \frac{1}{2} = 900 \quad B = (70 - 60) \cdot 60 = 600$$

$$C = (70 - 60) \cdot 20 \cdot \frac{1}{2} = 100 \quad D = (60 - 50) \cdot 60 = 600$$

$$E = (60 - 50) \cdot 20 \cdot \frac{1}{2} = 100 \quad F = (50 - 20) \cdot 60 \cdot \frac{1}{2} = 900$$

	CS	PS	DWL
Equilibrium	$A + B + C = 1600$	$D + E + F = 1600$	0
Ceiling at 50	$A + B + D = 2100$	$F = 900$	$C + E = 200$

$\Delta CS = 2100 - 1600 = +500$ and $\Delta PS = 900 - 1600 = -700$. Region D (600) is transferred from sellers to buyers. Regions $C + E$ (200) are lost to everyone, because the 20 bottles between 60 and 80 that were worth trading are no longer traded. Total surplus falls from 3200 to 3000.

Q3 | Price Floor

The Ministry of Magic decided that the market price wasn't sufficiently high enough for sellers, and decided to institute a price floor to raise the price. Use a graph of the demand curve and the supply curve to evaluate the welfare effects of a price floor of 70.

Quantity Supplied: _____

I 100 bottles.

Quantity Demanded: _____

I 60 bottles.

Excess, Market Clearing, or Shortage: _____

I Excess.

CS: _____

I 900 galleons.

How much did CS change? _____

I Down by 700 galleons.

PS: _____

I 2100 galleons.

How much did PS change? _____

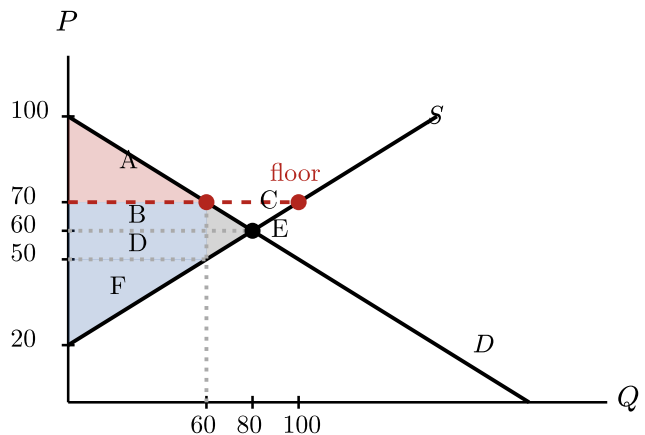
I Up by 500 galleons.

DWL: _____

I 200 galleons.

The floor of 70 sits above $P^* = 60$, so it binds. At 70: $Q_d = 60$ and $Q_s = 100$, an excess of 40 again only 60 bottles are exchanged, now at 70. The sellers who make the sales are the lowest-cost ones, whose sixtieth bottle costs $P = 20 + \frac{1}{2} \cdot 60 = 50$.

The same six regions from Q2 apply; only the price line moves from 50 up to 70. Buyers keep only A sellers collect B and D on top of F .



	CS	PS	DWL
Equilibrium	$A + B + C = 1600$	$D + E + F = 1600$	0
Floor at 70	$A = 900$	$B + D + F = 2100$	$C + E = 200$

$\Delta CS = 900 - 1600 = -700$ and $\Delta PS = 2100 - 1600 = +500$. Now B (600) is transferred from buyers to sellers, and the same $C + E$ (200) is lost. The ceiling and the floor are mirror images: each moves 600 from one side of the market to the other and destroys 200, because each stops the same 20 trades.